

Object detection

A detector predicts many candidate boxes and class scores; post-processing turns overlapping candidates into usable detections.

15–20 MINUTES

Prerequisite: CNN features

IOU EXERCISE

1 • Box

Approximate image location.

2 • Candidate

One possible object prediction.

3 • Score

Model evidence for a class.

4 • Post-process

Threshold and suppress duplicates.

Boxes are coordinate contracts

A box can be represented by corner coordinates (x_1, y_1, x_2, y_2) or center and size (cx, cy, w, h) . Values may be pixels or normalized fractions. A decoder must know which convention the model emits and which the consumer expects.

Central mental model

The network emits evidence-rich candidates. The decoder applies an artifact-specific contract and policy choices to produce final detections.

Why predict many candidates?

The number and locations of objects are unknown. One-stage detectors evaluate dense spatial positions and scales in one forward pass. Most candidates describe background or duplicates around the same object. A score threshold removes weak candidates; non-maximum suppression (NMS) removes redundant overlapping boxes.

A score is useful for ranking and thresholding under the model's training and calibration. It is not the observed frequency with which identical scores are correct unless calibration has been separately measured.

Intersection over Union

IoU measures overlap between two boxes:

$$\text{IoU} = \text{area}(\text{intersection}) / \text{area}(\text{union})$$

IoU is 0 for non-overlapping boxes and 1 for identical boxes. In NMS, keep a high-scoring candidate and suppress lower-scoring candidates of the same class whose IoU is above a chosen threshold. In evaluation, IoU can decide whether a prediction matches a ground-truth box. Those are two different uses and may use different thresholds.

Browser exercise: calculate overlap

Box A has area 100. Box B has area 100. Their intersection has area 60.

Calculate IoU, then reveal

If same-class NMS uses threshold 0.40, the lower-scoring box is suppressed. At 0.50, it is retained. Thresholds encode policy.

Applied to our cubes

VERIFIED ARTIFACT CONTRACT The current ONNX output is documented as `[1, 7, 8400]` : one batch item, seven fields, and 8,400 candidate positions. The fields are `cx, cy, w, h` plus three class scores.

DECODER BEHAVIOR The node transposes candidates, chooses the highest class score, applies a confidence mask, converts boxes, reverses letterboxing, clips coordinates, and runs per-class NMS.

BOUNDARY This layout describes the current export. Another YOLO generation, export option, or model can use a different contract.

Project anchors: model contract and decoder.

Check your understanding

What problem does NMS mainly address?

- ☐ Missing color channel values
- ☐ Duplicate overlapping object candidates
- ☐ Incorrect camera exposure settings

Explain it back

Trace one candidate from model output to final detection. Explain the different jobs of confidence thresholding, coordinate conversion, inverse letterboxing, and NMS.

Primary sources

- Original YOLO paper — unified one-stage detection idea.
- D2L: Bounding boxes and multiscale detection.